

TCC805x MCU BSP

API Specification for Fault Management Unit

Rev. 1.01 [A]

2021-01-22

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1 INTRODUCTION

This document describes the fault management unit device driver (hereinafter referred to as FMU driver) of TCC805x MCU BSP. For more detailed information of the fault management unit in the TCC805x MCU Sub-System, refer to Chapter 6.3.2 FMU (Fault Management Unit) of the *"PART 2-SMU_PMU"* in the *"TCC805x Full Specification"*. [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

SMU	System Management Unit
PMU	Power Management Unit
FMU	Fault Management Unit

3 SOURCE FILES

3.1 Location of Source Files

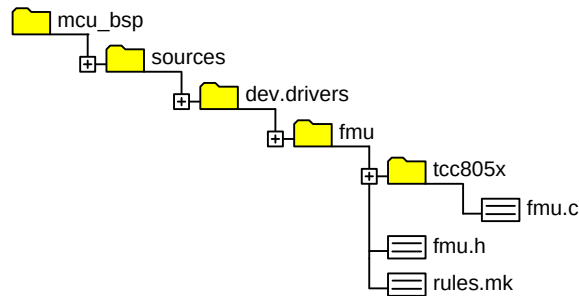


Figure 3.1 Location of Source Files

3.2 Simple Description of Source Files

- fmu.h
 - Defines external API and header information of the FMU driver.
- fmu.c
 - Supports setting and registering the interrupt for FMU so that it can detect a hardware fault and notify it to the outside of Cortex-R5 CPU or TCC805x.
- rules.mk
 - Rules used for build

4 ENUMERATIONS

4.1 FMUFaultid_t

Enumeration variables for fault source ID of FMU user interface API

Declaration

```
typedef enum FMUFaultid
{
    FMU_ID_ADC                = 0,
    FMU_ID_GPSB0              = 1,
    FMU_ID_GPSB1              = 2,
    FMU_ID_GPSB2              = 3,
    FMU_ID_GPSB3              = 4,
    FMU_ID_GPSB4              = 5,
    FMU_ID_GPSB5              = 6,
    FMU_ID_PDM0               = 7,
    FMU_ID_PDM1               = 8,
    FMU_ID_PDM2               = 9,
    FMU_ID_MC_CFG             = 10,
    FMU_ID_CR5_CFG            = 11,
    FMU_ID_CKC_CFG            = 12,
    FMU_ID_GPIO_CFG           = 13,
    FMU_ID_MID_CFG            = 14,
    FMU_ID_CMU_CFG            = 15,
    FMU_ID_SYS_SM_CFG         = 16,
    FMU_ID_X2X_MST0           = 17,
    FMU_ID_X2X_MST1           = 18,
    FMU_ID_X2X_SLV0           = 19,
    FMU_ID_X2X_SLV1           = 20,
    FMU_ID_X2X_SLV2           = 21,
    FMU_ID_SDM_VIDEO          = 22,
    FMU_ID_SDM_AUDIO          = 23,
    FMU_ID_SRVC               = 24,
    FMU_ID_PERI_CLK           = 25,
    FMU_ID_PLL1_CLK           = 26,
    FMU_ID_SRAM0_ECC          = 27,
    FMU_ID_SRAM1_ECC          = 28,
    FMU_ID_SFMC_ECC           = 29,
    FMU_ID_MBOX0_S_ECC        = 30,
    FMU_ID_MBOX0_NS_ECC       = 31,
    FMU_ID_MBOX1_S_ECC        = 32,
    FMU_ID_MBOX1_NS_ECC       = 33,
    FMU_ID_MBOX2_ECC          = 34,
    FMU_ID_OSC                 = 35,
    FMU_ID_VLD_CORE           = 36,
    FMU_ID_VLD_OTP            = 37,
    FMU_ID_WDT                 = 38,
    FMU_ID_PLL0_CLK           = 39,
    FMU_ID_CPU_CLK            = 40,
    FMU_ID_BUS_CLK            = 41,
    FMU_ID_CHIP_CFG_SOFT      = 42,
    FMU_ID_CHIP_CFG_CRC       = 43,
    FMU_ID_ROM_CRC            = 44,
```

```

    FMU_ID_FMU_CFG_SOFT    = 63
} FMUFaultId_t;

```

Description

Value	Description
<i>FMU_ID_ADC</i>	<i>ADC fault</i>
<i>FMU_ID_GPSB0</i>	<i>GPSB-0 fault</i>
<i>FMU_ID_GPSB1</i>	<i>GPSB-1 fault</i>
<i>FMU_ID_GPSB2</i>	<i>GPSB-2 fault</i>
<i>FMU_ID_GPSB3</i>	<i>GPSB-3 fault</i>
<i>FMU_ID_GPSB4</i>	<i>GPSB-4 fault</i>
<i>FMU_ID_GPSB5</i>	<i>GPSB-5 fault</i>
<i>FMU_ID_PDM0</i>	<i>PDM-0 fault</i>
<i>FMU_ID_PDM1</i>	<i>PDM-1 fault</i>
<i>FMU_ID_PDM2</i>	<i>PDM-2 fault</i>
<i>FMU_ID_MC_CFG</i>	<i>MCU configuration soft fault</i>
<i>FMU_ID_CR5_CFG</i>	<i>Cortex-R5 configuration soft fault</i>
<i>FMU_ID_CKC_CFG</i>	<i>Clock controller configuration soft fault</i>
<i>FMU_ID_GPIO_CFG</i>	<i>GPIO configuration soft fault</i>
<i>FMU_ID_MID_CFG</i>	<i>MID configuration soft fault</i>
<i>FMU_ID_CMU_CFG</i>	<i>Clock monitoring unit configuration soft fault</i>
<i>FMU_ID_SYS_SM_CFG</i>	<i>System safety mechanism configuration soft fault</i>
<i>FMU_ID_X2X_MST0</i>	<i>X2X master fault-0</i>
<i>FMU_ID_X2X_MST1</i>	<i>X2X master fault-1</i>
<i>FMU_ID_X2X_SLV0</i>	<i>X2X slave fault-0</i>
<i>FMU_ID_X2X_SLV1</i>	<i>X2X slave fault-1</i>
<i>FMU_ID_X2X_SLV2</i>	<i>X2X slave fault-2</i>
<i>FMU_ID_SDM_VIDEO</i>	<i>SDM fault-0</i>
<i>FMU_ID_SDM_AUDIO</i>	<i>SDM fault-1</i>
<i>FMU_ID_SRVC</i>	<i>SRVC fault</i>
<i>FMU_ID_PERI_CLK</i>	<i>Peripheral clock generator fault</i>

Value	Description
<i>FMU_ID_PLL1_CLK</i>	<i>PLL-1 output clock fault</i>
<i>FMU_ID_SRAM0_ECC</i>	<i>IMC RAM-0 ECC fault</i>
<i>FMU_ID_SRAM1_ECC</i>	<i>IMC RAM-1 ECC fault</i>
<i>FMU_ID_SFMC_ECC</i>	<i>SFMC RAM-1 ECC fault</i>
<i>FMU_ID_MBOX0_S_ECC</i>	<i>Secure MBOX-0 RAM ECC fault</i>
<i>FMU_ID_MBOX0_NS_ECC</i>	<i>Non-secure MBOX-0 RAM ECC fault</i>
<i>FMU_ID_MBOX1_S_ECC</i>	<i>Secure MBOX-1 RAM ECC fault</i>
<i>FMU_ID_MBOX1_NS_ECC</i>	<i>Non-secure MBOX-1 RAM ECC fault</i>
<i>FMU_ID_MBOX2_ECC</i>	<i>MBOX-2 RAM ECC fault</i>
<i>FMU_ID_OSC</i>	<i>XIN fault</i>
<i>FMU_ID_VLD_CORE</i>	<i>Core voltage fault</i>
<i>FMU_ID_VLD_OTP</i>	<i>OTPROM voltage fault</i>
<i>FMU_ID_WDT</i>	<i>Watchdog timer fault</i>
<i>FMU_ID_PLL0_CLK</i>	<i>PLL-0 output clock fault</i>
<i>FMU_ID_CPU_CLK</i>	<i>CPU clock fault</i>
<i>FMU_ID_BUS_CLK</i>	<i>Bus clock fault</i>
<i>FMU_ID_CHIP_CFG_SOFT</i>	<i>Chip configuration soft fault</i>
<i>FMU_ID_CHIP_CFG_CRC</i>	<i>Chip configuration CRC fault</i>
<i>FMU_ID_ROM_CRC</i>	<i>ROM CRC fault</i>
<i>FMU_ID_FMU_CFG_SOFT</i>	<i>FMU configuration soft fault (Only IRQ_EN/CLR/STS)</i>

Remark

None

4.2 FMUSeverityLevelType_t

Enumeration variables for FMU severity level types

Declaration

```
typedef enum FMUSeverityLevelType
{
    FMU_SVL_LOW    = 0,
    FMU_SVL_MID    = 1,
    FMU_SVL_HIGH   = 2
} FMUSeverityLevelType_t;
```

Description

Value	Description
FMU_SVL_LOW	The FMU generates an IRQ when a hardware fault occurs.
FMU_SVL_MID	The FMU generates an FIQ when a hardware fault occurs.
FMU_SVL_HIGH	The FMU generates an FIQ and External Error Indication Signal through GPIO-K[11] when a hardware fault occurs.

Remark

None

4.3 FMUErrTypes_t

Enumeration variables for FMU error types

Declaration

```
typedef enum FMUErrTypes
{
    FMU_OK                = 0,
    FMU_FAIL              = 1,
    FMU_INVALID_ID_ERR    = 2,
    FMU_RESERVE_ID_ERR    = 3,
    FMU_INVALID_SEVERITY_ERR = 4,
    FMU_ISR_HANDLER_ERR   = 5,
    FMU_DS_COMPARE_FAIL_ERR = 6,
    FMU_INTSET_ERR        = 7,
    FMU_INTEN_ERR         = 8,
    FMU_INTDIS_ERR        = 9
} FMUErrTypes_t;
```

Description

Value	Description
<i>FMU_OK</i>	<i>No error</i>
<i>FMU_FAIL</i>	<i>Unspecified FMU error occurs.</i>
<i>FMU_INVALID_ID_ERR</i>	<i>Invalid source fault ID value setting error</i>
<i>FMU_RESERVE_ID_ERR</i>	<i>Reserved source fault ID value setting error (45 ~ 62)</i>
<i>FMU_INVALID_SEVERITY_ERR</i>	<i>Invalid severity level value setting error</i>
<i>FMU_ISR_HANDLER_ERR</i>	<i>ISR Handler is NULL.</i>
<i>FMU_DS_COMPARE_FAIL_ERR</i>	<i>Double Storage result value is error.</i>
<i>FMU_INTSET_ERR</i>	<i>FMU interrupt setting error</i>
<i>FMU_INTEN_ERR</i>	<i>FMU interrupt enable error</i>
<i>FMU_INTDIS_ERR</i>	<i>FMU interrupt disable error</i>

Remark

None

5 CALLBACK FUNCTIONS

5.1 FMUIntFnctPtr

When a hardware error corresponding to a specific fault source ID occurs, the registered callback function is called and notified to the device driver.

Declaration

```
typedef void (*FMUIntFnctPtr)
(
    void*      pArg
);
```

Parameters

Value	Description
<i>pArg</i>	<i>[in/out] callback function argument</i>

Return

None

Remark

None

6 APIs

6.1 FMU_Init

FMU performs initialization as follows:

- Resets software of FMU.
- Sets the signal that notifies the hardware fault to the outside of TCC805x.
- Registers IRQ/FIQ interrupt handler.

Declaration

```
FMUErrTypes_t FMU_Init  
(  
    void  
);
```

Parameters

None

Return

Value	Description
FMUErrTypes_t	FMU Error type. Refer to the FMUErrTypes t .

Remark

None

6.2 FMU_IsrHandler

Stores the information of ID, severity level, interrupt handler callback function, and interrupt handler callback function parameters received from device drivers using FMU in global variable.

Declaration

```
FMUErrTypes_t FMU_IsrHandler
(
    FMUFaultid_t      uiId,
    FMUSeverityLevelType_t uiSeverityLevel,
    FMUIntFncPtr      fnIntFnc,
    void *             pArg
);
```

Parameters

Value	Description
<i>uiId</i>	[in] fault source ID. Refer to the FMUFaultid t .
<i>uiSeverityLevel</i>	[in] Severity Level. Refer to the FMUSeverityLevelType t .
<i>fnIntFnc</i>	[in] ISR Handler callback function. Refer to the <i>FMUIntFncPtr</i> .
<i>pArg</i>	[in] Argument

Return

Value	Description
<i>FMUErrTypes_t</i>	FMU Error type. Refer to the FMUErrTypes t .

Remark

None

6.3 FMU_Set

Sets the severity level, EN register, and IRQ/FIQ/Fault mask register to detect the hardware fault based on the received source ID from device drivers.

Declaration

```
FMUErrTypes_t FMU_Set  
(  
    FMUFaultid_t    uiId  
);
```

Parameters

Value	Description
uiId	[in] Fault source ID. Refer to the FMUFaultid t .

Return

Value	Description
FMUErrTypes_t	FMU Error type. Refer to the FMUErrTypes t .

Remark

None

6.4 FMU_IsrClr

The IRQ/FIQ/Fault clear register is set based on the severity level corresponding to the source ID received from the device drivers using FMU, so that the next interrupt can be received.

Declaration

```
FMUErrTypes_t FMU_IsrClr  
(  
    FMUFaultid_t    uiId  
);
```

Parameters

Value	Description
uiId	[in] Fault source ID. Refer to the FMUFaultid t .

Return

Value	Description
<i>FMUErrTypes_t</i>	<i>FMU Error type. Refer to the FMUErrTypes t.</i>

Remark

None

6.5 FMU_Deinit

Release the IRQ / FIQ interrupt handler registered in GIC400.

Declaration

```
FMUErrTypes_t FMU_Deinit  
(  
    void  
);
```

Parameters

None

Return

Value	Description
<i>FMUErrTypes_t</i>	<i>FMU Error type. Refer to the FMUErrTypes t.</i>

Remark

None

7 HOW TO USE FMU DRIVER

7.1 FMU Driver Initialization

Initialize the FMU driver by executing *fmu_init()*.

```
int BSP_Init ()
{
    I2C_Init();
    FMU_Init();           //FMU driver initialization
    GPSB_Init();
    ADC_Init(ADC_MODE_NORMAL);
}
```

7.2 Example of Using FMU in Other Device Drivers

The following example shows how to use FMU in device driver on GPSB device driver.

```
static void GPSB_FmuIsr0(void * pArg)
{
    FMUErrTypes_t ret = FMU_FAIL;

    ret = FMU_IsrClr(FMU_ID_GPSB0);           //interrupt clear

    return ret;
}

static int32 GPSB_EnableFmu (void)
{
    FMUErrTypes_t ret = FMU_FAIL;

    ret = FMU_IsrHandler(FMU_ID_GPSB0, FMU_SVL_LOW, GPSB_FmuIsr0, NULL); //register callback func
    ret = FMU_Set(FMU_ID_GPSB0);           //FMU set for HW fault

    return ret;
}
```

8 REFERENCES

- [1] TCC805x Full Specification
- [2] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 1.01: 2021-01-22

- Updated
 - Chapter 4 enumerations type define
- Deleted
 - Chapter 4.3 FMUFaultActive_t

Rev. 1.00: 2020-11-12

- First version release

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